Bonus Assignment: The Statistical Story

Introduction to Data Science · Optional bonus: up to 2 extra marks (marked out of 10 and scaled) · Set in Week 5 · Due at the start of the Week 9 class

Deadline: 30th November 2026 (After this date, NO ASSIGNMENT WILL BE ACCEPTED)

Note: You can either push it to your own repository and send me the link on WhatsApp or send me a ZIP file via email (a6dulrauf@gmail.com)

This assignment is optional. Whatever you earn adds to your total marks (the total never goes above 100) and can never lower your marks. It is the best place to practice the statistics from Weeks 5 and 6.

The situation

The head of department asks you: "What is going on with our final-exam results? Do the three sections differ? Does studying more lead to higher marks? Tell me what is true, what is uncertain, and what I should not conclude." You have one file and one week of statistics. Your job is to tell the story honestly.

The data

`exam_scores.csv`: 360 students in sections A, B, and C. Copy it from the data page or the **exam_scores.csv** that is given.

Column	Meaning
`student_id`	Anonymous ID
`section`	A, B or C
`study_hours_per_week`	Self-reported study time
`attendance_pct`	Percent of classes attended
`part_time_job`	Yes or No
`commute_minutes`	One-way travel time
`midterm_score`	Mark out of 100
`final_score`	Mark out of 100

The file has been cleaned already. You do not need to fix missing values, but you must still check what looks unusual.

What you must do

1. **Inspect and summarize.** Check the shape and the types. For each numeric variable, give the mean, median, standard deviation and IQR, and say which center you would report and why. Flag outliers in `commute_minutes` and `final_score` with the IQR rule and say whether each group looks like an error, a real but extreme value, or unknown.
2. **Compare the three sections on `final_score`.** Give the count, median, IQR and standard deviation of each. Draw one chart that shows the spread, not only the center. Say what differs and what does not.
3. **Study and scores.** Draw a scatter plot of `study_hours_per_week` against `final_score` and report the correlation. Then check whether `attendance_pct` could explain part of the relationship (for example, by comparing the correlation within attendance bands). Does studying more *cause* higher marks? Say what the data can and cannot show.
4. **Write the story** (one page): three findings, two limitations, and one misleading summary that you deliberately avoided and why.

What to submit

A notebook that runs from the top, with code and short written explanations, **three to four charts** (each with a title that states the finding), and the one-page story. Each chart must answer a question you state in the text.

How it is marked (10 marks)

Area	Marks	What earns full marks
Choosing and reading summaries, including outliers	3	Mean and median chosen to fit the shape (for example commute time is right-skewed); skew and spread interpreted; outliers flagged with the IQR rule and classified with a reason; nothing deleted without reason
Group comparison	3	Medians, IQR and counts shown; notices that one section is more spread out; the chart shows spread
Relationships and causation	2.5	Correlation reported and read correctly; checks attendance as a possible third factor; does not claim causation
The story and honesty	1.5	Clear findings with insight titles on the charts, genuine limitations, one misleading summary avoided

Rules

- Quote a center **and** a spread, never a center alone.
- A correlation is not a cause. Say what else could explain it.

- You may use an AI assistant as a copilot; attach the prompt log and verify every number against the data.

A note about what you will find

Not every variable behaves the same way, and not every difference is big. Do not assume that a difference in averages means a difference between groups. Look at the spread. Look at the counts. Say “we cannot tell” when that is the honest answer.